

How to use simulator

input

- IR_MAP.csv → Ri,Rp,C of target cell (see part-1 parameter identification)
- bat_pack_condition.csv → Initial conditions of battery pack (you can change with GUI)
- drive_condition.csv → Time-series output profile

src

- Batt_sim.py → Main file: just run this file
- bat_pack_cond_loader.py
- bat_sequential_calc.py
- interpolate_cell.py
- ir_map_load.py
- load_drive_condition.py

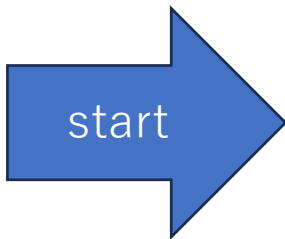
Place all of these files into a single folder. Please update the `csv\_path` in `Batt\_sim.py` to point to that folder's location.

you can change the input condition with this dialog

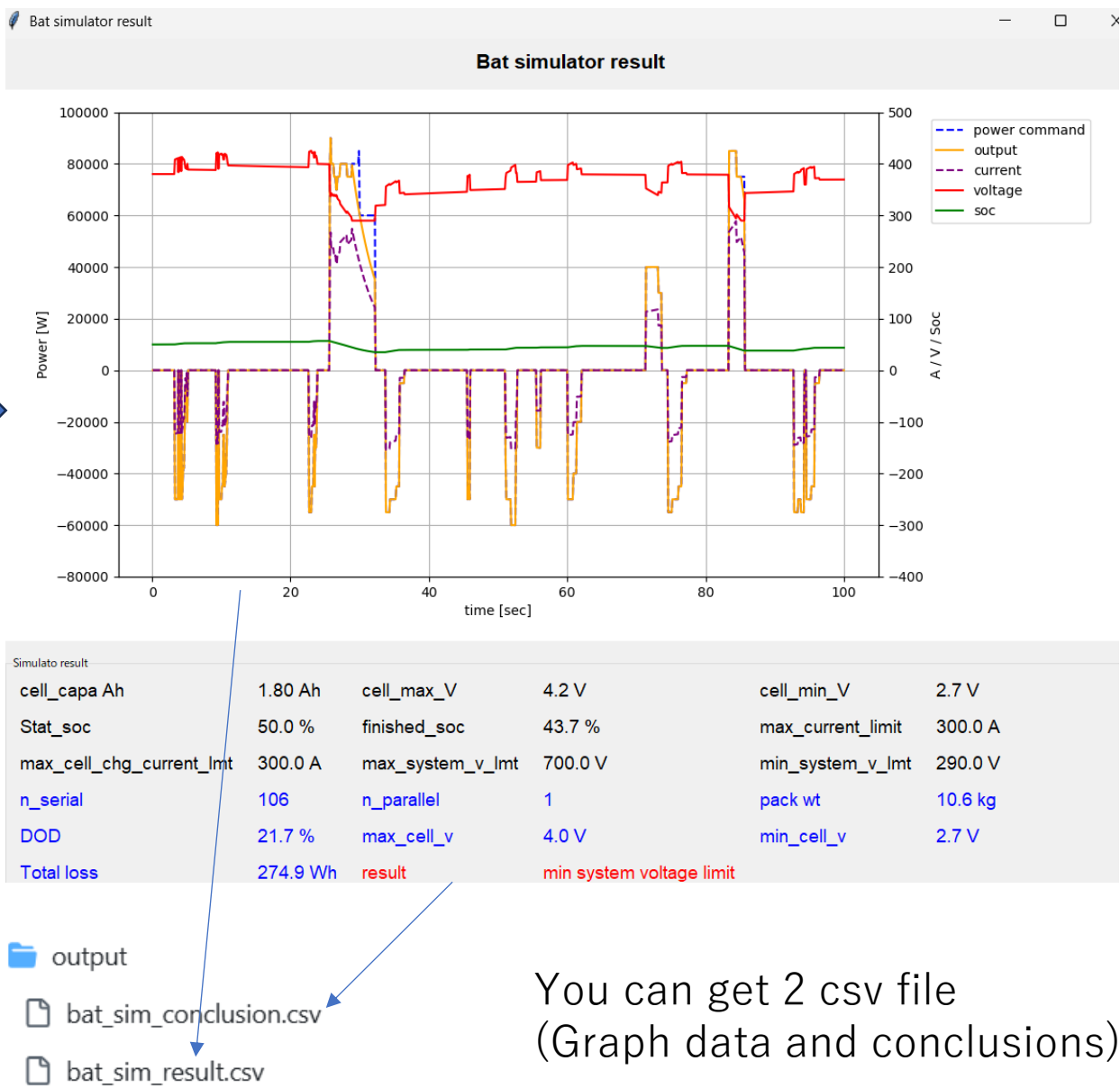
Battery pack condition

cell capa(Ah)	1.8
cell wt g	100.0
n-seri	106
n-para	1
start soc	50.0
system max current A	300.0
cell max charge current A	300.0
system max voltage V	700.0
system min voltage V	290.0
cell max lmt V	4.2
cell min lmt V	2.7

start simulation



output



You can get 2 csv file
(Graph data and conclusions)